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eHealth Digital Service Infrastructure
Electronic cross-border health services in the EU

Cryptographic Algorithms

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ABSTRACT
This normative binding defines constraints to the use of cryptographic algorithms and key lengths.

CHANGE HISTORY				
Version	Date	Status Changes	From	Review
V9.0.0	04/06/2025	Wave 9 Operation Ready: Update with SOGIS specifications v1.3	eHealth DSI Solution Provider	
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		document	eHealth DSI Solution Provider	
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V2.1.0	01/06/2017	Released for eHMSEG adoption	eHealth DSI Solution Provider	
V2.0.0	28/03/2017	Remove all references to epSOS and requirements	eHealth DSI provider	
V1.0	17/12/2013	Alignment with EED release documents	Fraunhofer FOKUS	

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1 Introduction

This normative binding defines constraints to the use of cryptographic algorithms and key lengths.

1.1 Related Documents

Requirements and logical level specifications related to this binding are provided in [Interoperability Specification].
eHealth DSI profiles for TLS certificates and for signature certificates as to be used for signing SAML assertions are provided in [X509 Certificate Profiles].

1.2 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].
eHealth DSI related terms (e.g. NCPeH) are be interpreted as defined in [Interoperability Specification] and [eHealth DSI Glossary].

1.3 Status of this Binding

The binding as defined in this document is a normative binding. All eHealth DSI countries MUST implement this binding within their NCP.
Additional or alternative bindings MUST NOT be defined and implemented for the eHealth DSI system.

2 Cryptographic Keys and Algorithms

All cryptographic keys and algorithms used for eHealth DSI and its implementations MUST fulfil at least the requirements of [SOG-IS-2020] for "Legacy" level of security with validity longer than end of 2025. The minimum key size for a block cipher (symmetric encryption) MUST be 128 bits.

The use of the "Recommended" or stronger cryptographic keys and algorithms defined in [SOG-IS-2020] is recommended (SHOULD).

[SOG-IS-2020] recommendations define the eHealth DSI minimum requirements on the selection of cryptographic keys and algorithms.

Algorithms based on elliptic curves MAY be used if agreed by all countries that participate in eHealth DSI.

3 Transport Layer Security

Transport Layer Security (TLS) and its now-deprecated predecessor, Secure Sockets Layer (SSL) are protocols that provide authentication, privacy, and data integrity between two communicating computer applications.
It's most prominently used to secure the data that travels between a web browser and website via HTTPS, but it can also be used to secure web services and a host of other protocols.
SSL 2.0, SSL 3.0, TLS 1.0 and TLS 1.1 are now considered deprecated, but version 1.2 is in widespread use. Version 1.3 is starting to see increased adoption.
In the eHDSI Trust Model, at least TLS 1.2 SHALL be used. TLS 1.3 is RECOMMENDED.

The choice of cipher suites MUST follow [SOG-IS-2020, p 32]. CipherSuites being not longer valid than end of 2025 MUST NOT be used." chapter SOG-IS-2020#6 "Cryptographic Protocols.

4 XML Digital Signatures and XML Encryption

4.1 Assertion Signature

Every SAML Assertion that is used for brokering identity and context information among NCPs MUST be signed by its issuer. The XML signature MUST be applied by using the *saml:Assertion/ds:Signature* element as defined below.

Signature Parameter	Usage Convention
CanonicalizationMethod	SHOULD be "http://www.w3.org/2001/10/xml-exc-c14n#"
Transformation	Enveloped signature transform according to section 6.6.4 of [W3C XMLDSig] SHOULD be used http://www.w3.org/2000/09/xmldsig#enveloped-signature. In addition, exclusive canonicalization SHOULD be defined as transformation (http://www.w3.org/2001/10/xml-exc-c14n#, according. [W3C XMLDSig] and [W3C XML-EXC 1.0]). As inclusive namespaces other prefixes than the ones defined in [[SAML Profile]] MUST NOT be used.
SignatureMethod	The signature method MUST comply with the eHealth DSI recommendations on algorithms and key lengths (see chapter 2). For signing assertions the signature method • http://www.w3.org/2001/04/xmldsig-more#rsa-sha256 SHOULD be used.
DigestMethod	The hash algorithm MUST comply with the eHealth DSI recommendations on algorithms and key lengths (see chapter 2). For signing assertions the digest method • http://www.w3.org/2001/04/xmlenc#sha256 SHOULD be used.
KeyInfo	This element MUST either contain a wsse:SecurityTokenReference element which references the X.509 certificate of the assertion's issuer by using a subject key identifier OR contain a ds:X509 Data element which contains the X.509 certificate of the assertion issuer.

4.2 *XML ENCRYPTION*

XML Encryption has shown to be vulnerable against man-in-the-middle attacks. Therefore XML encrypted data MUST only be shared among eHealth DSI actors through encrypted transportation channels.

5 References

5.1 Normative References

[SOG-IS-2020] SOG-IS Crypto Evaluation Scheme. Agreed Cryptographic Mechanisms. Version 1.4. January 2020. https://www.sogis.eu/documents/cc/crypto/SOGIS-Agreed-Cryptographic-Mechanisms-1.3.pdf

[RFC 2119] Bradner, S.: Key words for use in RFCs to Indicate Requirement Levels.
Harvard University, Boston, Massachusetts, 1997.

[W3C XMLDSig] W3C: XML Signature Syntax and Processing (Second Edition). W3C Recommendation, June 2008.
http://www.w3.org/TR/xmldsig-core/

[W3C XML-EXC 1.0] W3C: Exclusive XML Canonicalization Version 1.0. W3C Recommendation, July 2002.
https://www.w3.org/TR/xml-exc-c14n/